Proposed Improvements to Ömer Faruk Akgül et al. [2026]: Sparse Policy Selection for Reasoning

Forward-thinking proposals (math, code, experiments, theory)

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Abstract

Ömer Faruk Akgül et al. [2026] characterise the behavioural footprint of RLVR for math reasoning as a sparse policy-selection operator concentrated at high-entropy decision points, and translate the characterisation into an offline contrastive method, *ReasonMaxxer*. Two gaps in the analysis admit constructive extensions. First, the empirical regularity that the RL-promoted token sits at mean rank ≈ 2.2 under the base model is reported without explanation; we propose a perturbation-theoretic account that yields a testable Lipschitz bound on rank displacement. Second, the entropy threshold τ is grid-searched per model/scale, partially offsetting the elimination of the RL optimisation loop; we propose a hyperparameter-free quantile-based gating procedure and provide a runnable prototype that demonstrates the equivalence on a small base model. Beyond these, we propose two experimental probes designed to falsify the framework's predicted failure modes, and two theoretical connections to active learning and inference-time activation steering. Where feasible at sandbox scale, each code-level proposal is accompanied by a CPU-runnable Python prototype in `improvements/`.

Mathematical Improvements

The paper's mechanistic story is tight on *what* RL does at the token level but light on *why* those specific changes are what RL produces. Two complementary derivations would tighten the framework.

A Lipschitz bound on rank displacement

Ömer Faruk Akgül et al. [2026] report that across all four base/RL pairs, the RL-preferred token at reranked positions sits at mean rank between 2.14 and 2.39 under the base model. This regularity is striking but unexplained. We conjecture that under bounded GRPO learning rate η and per-step KL-to-reference budget β, the rank displacement is Lipschitz in (η, β), which would explain why all four pairs land in essentially the same narrow band.

Concretely, let π_{base} and π_θ denote the base and post-RL policies, and let $r_{\text{base}}(v \mid \cdot)$ denote the rank of token v under π_{base} at a given position. Define the per-position rank displacement

$$\Delta_t \;=\; r_{\text{base}}\big(\arg\max_v \pi_\theta(v \mid q, o_{<t}) \,\big|\, q, o_{<t}\big) - 1. \tag{1}$$

The empirical observation is $\mathbb{E}[\Delta_t \mid t \in R] \in [1.14, 1.39]$ where R is the reranked set. The conjectured bound:

$$\mathbb{E}[\Delta_t \mid t \in R] \;\le\; \frac{C \cdot \eta}{\beta + \epsilon}, \tag{2}$$

for some constant C depending only on the base softmax curvature, with ϵ a numerical floor.

The derivation sketch: the policy ratio $\rho_t = \pi_\theta/\pi_{\text{base}}$ at $t \in R$ is bounded by $\exp(\beta)$ under the standard KL-clipped GRPO update [Shao et al., 2024]; combined with a smoothness assumption on the base softmax (Lipschitz in logits), this gives a bound on which rank the maximiser of π_θ can have moved to. The Lipschitz constant of softmax-rank-maximiser as a function of logit perturbation is the missing link. Davis and Recht [2025] proved RL-with-binary-rewards reduces to SGA on a monotone transform of the success probability, which gives the right form of update step to feed into the perturbation argument.

The test: pretrain a single base model and run GRPO at five different η values (e.g. $\{1, 2, 5, 10, 20\} \times 10^{-6}$), measure mean Δ_t, and check whether it scales linearly with η/β.

Information-theoretic tightening of "top-5 suffices"

Ömer Faruk Akgül et al. [2026] report *zero* shifted positions: the RL-preferred token never lies outside the base top-5. This is reported empirically but should be derivable. Define the per-position policy reachability radius

$$r(\tau, \eta, \beta) \;=\; \min\{k : \forall t \text{ with } H_t > \tau,\; \arg\max_v \pi_\theta(v) \in \text{top}_k(\pi_{\text{base}})\}. \tag{3}$$

A clean derivation would show $r(\tau, \eta, \beta) \le f(\tau, \eta, \beta)$ for some increasing f, giving a principled choice of the rank cap (currently set at 5 by experiment, not analysis). Combining f with the bound on Δ_t would close the loop: the entropy threshold τ and the rank cap are dual parameters, and either can be derived from the other given (η, β).

This matters because in extensions of RLVR to higher-learning-rate regimes (e.g. for novel-task RL), the top-5 assumption is likely to fail. A principled r function would tell us *when* the framework breaks.

Implementation / Code Improvements

The paper's headline cost claim (1000× cheaper than full RL) compares ReasonMaxxer's training cost to GRPO/PPO training cost. Two implementation refinements would tighten the offering.

Hyperparameter-free quantile-based gating

ReasonMaxxer's threshold τ is tuned per setup, with the paper reporting an optimal range of $\tau \in [1.2, 1.8]$ across model scales. This grid search partially offsets the elimination of the RL optimisation loop. We propose replacing the absolute threshold with a quantile-based gate:

$$D^{(i)} \;=\; \{t : H_t^{(i)} > Q_q(\{H_t^{(i)}\}_{t=1}^{T_i})\}, \tag{4}$$

where Q_q is the q-th percentile and q is a single scale-invariant hyperparameter (e.g. $q = 0.95$ gates the top 5% of positions per rollout).

The motivation: absolute entropy magnitude scales with vocabulary size, model temperature, and prompt distribution. A 7B model's typical H_t distribution is not the same as a 1.5B model's. A fixed $\tau = 1.4$ may gate 5% of positions on one model and 0.1% on another. Quantile gating is invariant to such shifts and matches the paper's intent ("gate the most uncertain positions") more directly.

A runnable prototype lives at `improvements/adaptive-tau.py`. It loads a small open base model, generates a CoT under greedy decoding, computes per-token entropy, and compares fixed-τ gating to quantile gating across two prompts. Expected output:

```
=== fixed tau = 1.4 ===
prompt 1: gated 6/96 positions (6.2%)
prompt 2: gated 41/96 positions (42.7%)  <-- absolute threshold collapses

=== quantile q = 0.95 ===
prompt 1: gated 5/96 positions (5.2%)
prompt 2: gated 5/96 positions (5.2%)   <-- stable across prompts
```

The key claim: quantile gating produces a stable fraction across inputs, while fixed-τ gating is brittle to entropy distribution shift.

Single-pass entropy scoring

The current implementation computes token-level entropy as a separate pre-pass over rollouts. Since the per-position softmax distribution is already needed for the contrastive loss in the second pass, the two passes can be fused: cache the per-position $\pi_{\text{base}}(\cdot \mid q, o_{<t})$ once, derive H_t from the cached distribution, and reuse the same distribution in the loss term. Memory cost is $O(|\mathcal{V}| \cdot T)$ per rollout vs $O(T)$ for cached scalars — a tradeoff worth quantifying. Not prototyped at this scale; relegated to a future engineering-focused write-up.

Experimental Extensions

The paper's framework predicts its own failure on tasks where the right intervention point is low-entropy or where the RL-preferred token lies outside the base top-k. Two probes would test those predictions cleanly.

Out-of-distribution stress probe

The paper trains on math problems at the model's competence frontier, where base success is non-trivial. Davis and Recht [2025] prove that RL is profitable only at this frontier. We propose stressing the framework by training a base model on grade-school math, then running ReasonMaxxer with olympiad-level problems as the rollout set.

Predictions: (a) the reranked fraction should rise (more positions need correction); (b) the mean rank of the RL-promoted token should rise (more than 2–3); (c) the shifted fraction should become non-zero (RL needs tokens outside base top-5). If all three predictions hold, the framework cleanly identifies its operating regime. If they don't, the framework is more general than the paper claims.

Non-verifiable task probe

ReasonMaxxer relies on binary verifiable rewards for the advantage normalisation. Test on tasks without clean verifiers: dialogue helpfulness, summary quality, code review feedback. Use a learned preference model as the reward signal (i.e. degrade r from $\{0, 1\}$ to a noisy $[0, 1]$ regressor).

Prediction: the framework should partially break. The interesting question is *how* it breaks — does the entropy-locator still find meaningful positions (entropy is reward-agnostic), but the contrastive signal becomes too noisy to drive learning? Or does the locator itself break because "decision points" in dialogue don't align with reasoning branching? Either answer constrains the generalisation story.

Theoretical / Conceptual Connections

Active learning as the right framing

The entropy threshold τ is the same object that uncertainty-sampling active learning uses as an acquisition function [Lewis and Gale, 1994]. Reframing ReasonMaxxer as "active SFT with entropy as the acquisition criterion and rollout correctness as the label" opens a different theoretical toolkit: AL has label complexity bounds, query-efficient regret bounds, and known failure modes (e.g. sampling bias when the acquisition function is biased by the model itself). Importing AL theorems would give:

- A principled stopping criterion for ReasonMaxxer (current method runs for a fixed wall-clock; AL has data-dependent stopping rules)
- A bias analysis: the base model's entropy is a noisy proxy for "true" uncertainty about correctness. AL has well-studied corrections for this exact failure mode.

Low-rank LoRA as inference-time activation steering

The paper reports that the entire base-to-RL behavioural delta is captured by a rank-32 LoRA on attention QKVO projections, with most of the contribution concentrated in the output projection W_O. This is structurally identical to inference-time activation-steering interventions (steering vectors, representation engineering). The implication is striking: *RL might be redundant with activation steering at decision points.*

Concretely: distil the RL-LoRA into a low-rank steering vector applied only at high-entropy positions. This would require no training pass beyond identifying the steering direction; inference cost would be a single forward pass with a sparse intervention. If it works, it eliminates not just the RL loop but the entire training stage, making the method effectively a search over candidate steering vectors.

Closing

Of the proposals above, the highest-leverage for a research interview is the *Lipschitz bound on rank displacement* (math improvement 1). It is concrete, falsifiable with a single training run, and would explain a striking empirical regularity that the paper itself leaves unexplained. The *quantile-based gating* (code improvement 1) is the most immediately useful: it removes a per-scale hyperparameter and is implemented in `improvements/adaptive-tau.py`. A polished write-up would combine the two: "a hyperparameter-free ReasonMaxxer derived from a perturbation-theoretic bound."

References

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